

Computing the Centre of Elasticity in Planar Frames

Vishwanath Ravindran^{1,*} and G. K. Ananthasuresh¹

¹Department of Mechanical Engineering, Indian Institute of Science, Bengaluru, India
*vishwanathr@iisc.ac.in

Abstract. At the centre of elasticity, the rotational and translational degrees of freedom are decoupled, i.e. linear forces applied at this point only cause translational displacement, and torque applied at this point causes only rotational displacement. This work focuses on the computation of the centre of elasticity of planar frames. The procedure to find the centre of elasticity of a cantilever beam is detailed, along with the theory on serial and parallel concatenation of frame elements. The theory is presented here in terms of only the stiffness or compliance matrices instead of the screw theory-based approach usually adopted in the literature. The goal of this work is to provide intuition for the centre of elasticity for design practitioners. Interestingly, it is shown that the centre of elasticity about the terminal node of serially assembled beams of uniform flexural rigidity coincides with its centroid.

Keywords: Compliant mechanisms, Centre of elasticity, Stiffness ellipsoid, Compliance ellipsoid

1 Introduction

The centre of elasticity (CoE) refers to a point where the application of translational forces results in pure translational displacement of the point (without any rotational displacement of the point), and, equivalently, the point undergoes pure rotational displacement (without translational displacement) under the application of a pure moment about that point (for small deformation). Essentially, the rotational and translational components of the compliance matrix are decoupled at this point. The centre of elasticity does not refer to any material point in a continuum body, but instead, it refers to a spatial coordinate defined for each material point in the continuum body. The centre of elasticity concept maps each material point to a particular spatial coordinate, and if this point were to be connected with a rigid link, the translation and rotation about this particular spatial coordinate would be decoupled. The existence of such a point is guaranteed for each material point of a 2D continuum body (see section 2.1).

In [1], Krishnan and coworkers had adopted a representation based on the centre of elasticity to decompose a point stiffness matrix in terms of a translational compliance ellipse, eigen-rotational stiffness, and a coupling vector. The last two are related to the centre of elasticity corresponding to that point. They studied the serial and parallel concatenation of compliant elements (building blocks) in this representation. They had then specialised their relations for a compliant dyad. These were used to derive design charts, which can be used

to engineer an arbitrary 2D point stiffness or compliance matrix. There are also other mathematically rigorous investigations on these topics based on screw theory [2] or ellipse of elasticity theory [3]. However, the mathematical rigour and concise representation make it challenging for design practitioners to adopt these methods.

To this end, the study on the centre of elasticity for a frame element is presented here with only the stiffness and compliance matrix without any abstract notation. This is done by first deriving the coordinates of the centre of elasticity based on the stiffness and compliance matrix about the point of interest (PoI). Using this, the centre of elasticity for a cantilever beam is calculated. Then, the relations for computing the centre of elasticity of a device constructed by serial and parallel concatenation of individual compliant elements with known centres of elasticity are derived. Using these results, it is possible, in principle, to calculate the centre of elasticity of any frame element by decomposing it as a serial and parallel concatenation of beam elements.

2 Stiffness and Compliance matrices under Rigid-body Transformations

An illustrative schematic of the rigid-body transformation of PoI, along with adopted notations, is presented in Fig. 1. In general, it is possible to represent the transformation matrix as a rigid-body rotation followed by a rigid-body translation. Rigid-body translation essentially means that the PoI is connected to the translated coordinate by a rigid-body, and rigid-body rotation refers to rotation of the basis about the point¹. All the matrices and vectors can be represented about this new coordinate basis. The basis transformation for forces and displacements about the PoI are as follows²:

$$\begin{Bmatrix} F_x^1 \\ F_y^1 \\ M_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta + x \sin \beta & x \cos \beta + y \sin \beta & 1 \end{bmatrix} \begin{Bmatrix} F_x \\ F_y \\ M_z \end{Bmatrix}, \quad (1)$$

$$\begin{Bmatrix} u_x^1 \\ v_y^1 \\ \theta_z^1 \end{Bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta & y \\ \sin \beta & \cos \beta & -x \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_x \\ v_y \\ \theta_z \end{Bmatrix} \quad (2)$$

Then, the new stiffness matrix³ ($[\mathbf{K}]_{3 \times 3}$) after transformation can be related to the stiffness matrix about PoI ($[\mathbf{K}^1]_{3 \times 3}$) as shown below:

$$[\mathbf{K}] = \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ -y \cos \beta + x \sin \beta & x \cos \beta + y \sin \beta & 1 \end{bmatrix}^{-1} [\mathbf{K}^1] \begin{bmatrix} \cos \beta & -\sin \beta & y \\ \sin \beta & \cos \beta & -x \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

¹ Origin is considered to coincide with the PoI.

² forces about all degrees of freedom other than that associated with the PoI are considered absent

³ In this work, stiffness matrix refers to the single-point stiffness matrix constructed by inverting the single-point compliance matrix ($[\mathbf{C}]_{3 \times 3}$: such that $[\mathbf{C}] \{\mathbf{f}\} = \{\mathbf{u}\}$).

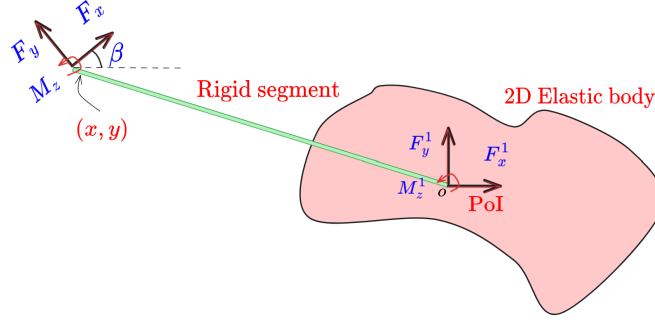


Fig. 1: Illustration of rigid-body transformation of PoI

2.1 Rigid-body translation of point of interest

First, considering $\beta = 0$ (rigid-body translation)

$$[\mathbf{K}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y & x & 1 \end{bmatrix}^{-1} [\mathbf{K}^1] \begin{bmatrix} 1 & 0 & y \\ 0 & 1 & -x \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} K_{11}^1 & K_{12}^1 & K_{31}^1 - x(K_{12}^1) + y(K_{11}^1) \\ \cdot & K_{22}^1 & K_{23}^1 - x(K_{22}^1) + y(K_{12}^1) \\ \cdot & \cdot & K_{33}^1 + (K_{11}^1)y^2 + (K_{22}^1)x^2 - (2K_{12}^1)xy + (2K_{13}^1)y - (2K_{23}^1)x \end{bmatrix} \quad (4)$$

Similarly, the compliance matrix undergoes the following transformation:

$$[\mathbf{C}] = \begin{bmatrix} 1 & 0 & y \\ 0 & 1 & -x \\ 0 & 0 & 1 \end{bmatrix}^{-1} [\mathbf{C}^1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -y & x & 1 \end{bmatrix} = \begin{bmatrix} C_{11}^1 - (2C_{13}^1)y + C_{33}^1y^2 & \cdot & \cdot \\ C_{12}^1 + (C_{13}^1)x - (C_{23}^1)y - (C_{33}^1)xy & (C_{33}^1)x^2 + (2C_{23}^1)x + C_{22}^1 & \cdot \\ C_{13}^1 - (C_{33}^1)y & C_{23}^1 + (C_{33}^1)x & C_{33}^1 \end{bmatrix} \quad (5)$$

From the preceding exercise, it can be observed that the translation components of the stiffness matrix (K_{11} , K_{12} , K_{22}) and the compliance associated with the rotational degree of freedom (C_{33}) remain unaffected upon rigid-body translation. Also, the centre of elasticity can be computed by equating K_{31} and K_{32} or equivalently C_{31} and C_{32} to zero. The latter leads to the representation of the coordinates of the centre of elasticity ($\{x_c, y_c\} = \{-C_{32}^1/C_{33}^1, C_{31}^1/C_{33}^1\}$) in terms of the coupling vector ($\{C_{31}^1, C_{32}^1\}^T$) and the compliance associated with the rotational degree of freedom (C_{33}). Thus, rigid-body translation is sufficient to decouple the translation and rotational degrees of freedom.

2.2 Rigid-body rotation of point of interest

Now, for rigid-body rotation ($\{x, y\} = \{0, 0\}$), the general expression Eq. (3) simplifies to the following form:

$$\begin{aligned}
 [\mathbf{K}] &= \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} [\mathbf{K}^1] \begin{bmatrix} \cos \beta & -\sin \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} K_{11} & K_{12} & K_{13}^1 \cos(\beta) + K_{23}^1 \sin(\beta) \\ . & K_{22} & K_{23}^1 \cos(\beta) - K_{13}^1 \sin(\beta) \\ . & . & K_{33}^1 \end{bmatrix}
 \end{aligned} \tag{6}$$

The translation components ($K_{11} = K_{11}^1 \cos^2(\beta) + K_{22}^1 \sin^2(\beta) + K_{12}^1 \sin(2\beta)$, $K_{12} = K_{12}^1 \cos(2\beta) + 0.5(K_{22}^1 - K_{11}^1) \sin(2\beta)$, $K_{22} = K_{22}^1 \cos^2(\beta) + K_{11}^1 \sin^2(\beta) - K_{12}^1 \sin(2\beta)$) of the new stiffness matrix are derived from Mohr circle transformation of translation components of the old stiffness matrix ($K_{11}^1, K_{12}^1, K_{22}^1$). The compliance matrix undergoes the same transformation upon rigid-body rotation. The rotation operation can be performed to decouple the translational components along orthogonal directions (by setting K_{12} to zero). Also, the compliance associated with the rotational degree of freedom (C_{33}) is unaffected by both rigid-body rotation and translation.

3 Computing the Centre of Elasticity in a Frame

3.1 Centre of elasticity of a cantilever beam

The procedure for computing the centre of elasticity of a cantilever beam is detailed here. The stiffness matrix of a cantilever beam is shown below:

$$[\mathbf{K}^1] = \begin{bmatrix} EA/L & 0 & 0 \\ 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & -6EI/L^2 & 4EI/L \end{bmatrix}, \tag{7}$$

where the symbols have their usual meaning. The eigenvalues and eigenvectors of this stiffness matrix are shown below:

$$\begin{aligned}
 \Rightarrow [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2L}{3} - \frac{\sigma_2 + \sigma_1 + EI L^2}{L \sigma_2} & \frac{2L}{3} - \frac{\sigma_2 - \sigma_1 + EI L^2}{L \sigma_2} \\ 0 & 1 & 1 \end{bmatrix} \text{ and} \\
 \{\lambda_1, \lambda_2, \lambda_3\} &= \begin{bmatrix} \frac{AE}{L} & \frac{2(3EI + \sigma_1 + EI L^2)}{L^3} & \frac{2(3EI - \sigma_1 + EI L^2)}{L^3} \end{bmatrix}
 \end{aligned} \tag{8}$$

where, $\sigma_1 = EI\sqrt{L^4 + 3L^2 + 9}$ and $\sigma_2 = 3EI$.

It should be noted that the calculations involve dimensionally inconsistent operations. This makes the interpretation of these relations difficult. One way to address this is by introducing normalising lengths [4]. Which also allows the

construction of physically interpretable compliance ellipsoids based on the eigenvalues and eigenvectors. But deciding on a normalising length scale is not immediately obvious. Instead, the eigen-wrench and eigen-twist representation based on screw theory is adopted commonly, which allows for a mathematically rigorous and natural way to decouple the translational and rotational components. Also, it is critical to note that although the stiffness matrix (due to its symmetry) can be diagonalised by rotational basis transformation using the eigenvectors in a manner similar to Cauchy stress or inertia tensor, such rotational basis transformation is not admissible in general as the transformation matrix must take the form expressed in Eq. (6). And the normalised eigenvectors (from Eq. (8)) cannot be represented in this form. Thus, rotational basis transformation alone will not suffice to decouple the translational and rotational motion. Thus, there is a need to perform basis translation.

Now, the CoE ($\{x_c, y_c\}$) can be found by translating the PoI to the CoE and setting $K_{23} = K_{32} = 0$, $K_{13} = K_{31} = 0$. This leads to ($\{x_c = L/2, y_c = 0\}$).

$$[\mathbf{K}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -L/2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} EA/L & 0 & 0 \\ 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & L/2 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

$$= \begin{bmatrix} AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 0 \\ 0 & 0 & EI/L \end{bmatrix}$$

Essentially, the centre of elasticity of a cantilever beam is coincident with the centre of the beam. Decoupling of translation and rotation about this point is shown in Fig. 2.

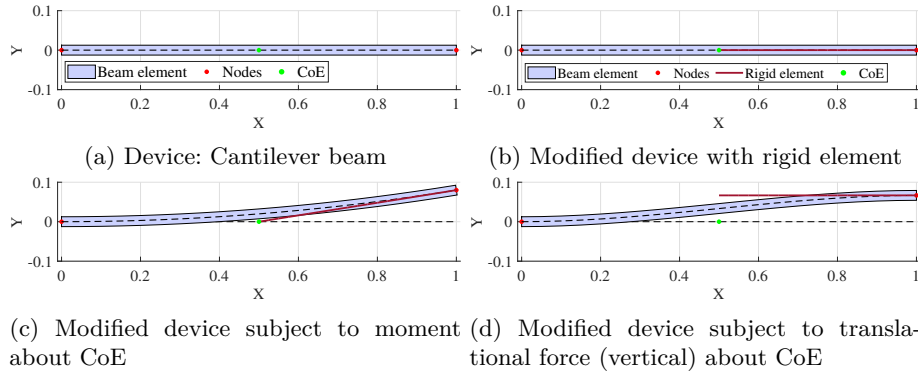


Fig. 2: Illustration of CoE concept for a cantilever beam

3.2 Serial concatenation of elements

The methodology used to compute the centre of elasticity for the serial composition of elements is visually depicted in Fig. 3. Let the single-point stiffness matrix of element-1 about CoE₁ ($\{x_1, y_1\}$) and element-2 about CoE₂ ($\{x_2, y_2\}$) be given by:

$$[\mathbf{C}_1] = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{33} \end{bmatrix}, \quad [\mathbf{C}_2] = \begin{bmatrix} G_{11} & G_{12} & 0 \\ G_{12} & G_{22} & 0 \\ 0 & 0 & G_{33} \end{bmatrix} \quad (10)$$

First, the PoI of the element's compliance matrix is translated from their respective CoE to the loading tip of element-2 (t). The net compliance matrix is merely their sum:

$$[\mathbf{C}_t^{net}] = \begin{bmatrix} C_{11} + G_{11} & C_{12} + G_{12} & -C_{33}(y_t - y_1) - G_{33}(y_t - y_2) \\ \cdot & C_{22} + G_{22} & C_{33}(x_t - x_1) + G_{33}(x_t - x_2) \\ \cdot & \cdot & C_{33} + G_{33} \end{bmatrix} \quad (11)$$

Now, based on arguments made earlier regarding Eq. (5), the centre of elasticity for the concatenated device is

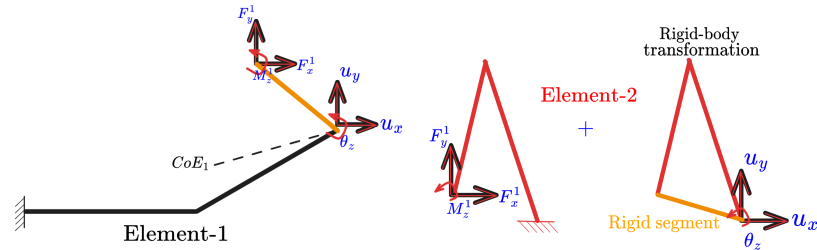
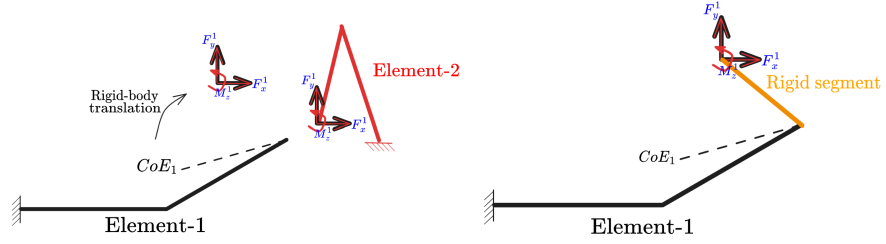
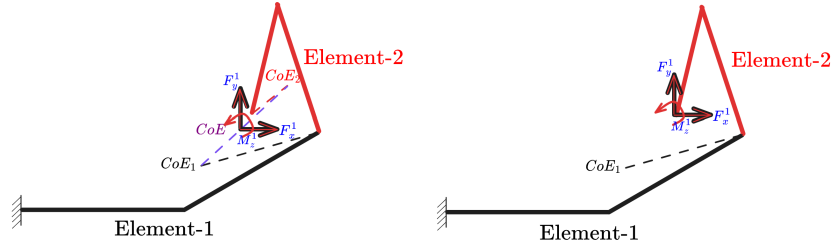
$$y_{\text{CoE}} = \frac{(C_{33})y_1 + (G_{33})y_2}{C_{33} + G_{33}}, \quad x_{\text{CoE}} = \frac{(C_{33})x_1 + (G_{33})x_2}{C_{33} + G_{33}} \quad (12)$$

The above expression leads to a special property when specialised for beam elements. For a beam element, the eigen-rotational compliance ($C_{33}|_{\text{CoE}}$) is $L/(EI)$, and the centre of elasticity is the midpoint of the beam. Thus, for serial concatenation of n beam elements, the position of the centre of elasticity is:

$$\mathbf{r}_{\text{CoE}} = \frac{\sum_{i=1}^n (L_i/E_i I_i) \mathbf{r}_{m,i}}{\sum_{i=1}^n (L_i/E_i I_i)}, \quad (13)$$

where $\mathbf{r}_m$ refers to the position of the midpoint of each beam. In case the members have uniform flexural rigidity, the centre of elasticity about the terminal node is coincident with the centroid of the frame. Examples illustrating this fact is shown in Fig. 4. For the device shown in Fig. 4a, a new device constructed by rigid-body translation of PoI to the centre of elasticity is shown in Fig. 4b. It is also possible to construct a new device for which the CoE is coincident with PoI through the use of a compliant link as shown in Fig. 4e (the compliant element shares the same properties as rest of the members) by suitably accounting for the compliance of the new element in Eq. (13) (in this case the CoE lies in the line joining the centroid and terminal node of the original device).

Thus, for a serial concatenation of beam elements with uniform flexural rigidity, the centre of elasticity (centroid) will always be within the footprint of the device. This is also true when the beam elements have different flexural rigidity as shown mathematically in [1]. Also, it is not always possible to realise these devices in 2D, due to the self-intersection of beams in the construction of the modified device (as evident in Fig. 4i). This can be overlooked, as in practice, it is possible to construct this device by slightly elevating the rigid element out of the plane.



(f) The displacements of the PoI of element-2, due to deformation of element-2 are superimposed with the displacement at the same point due to deformation of element-1

Fig. 3: Serial concatenation of representative elements, CoE of resultant device lies in line joining CoE_1 and CoE_2

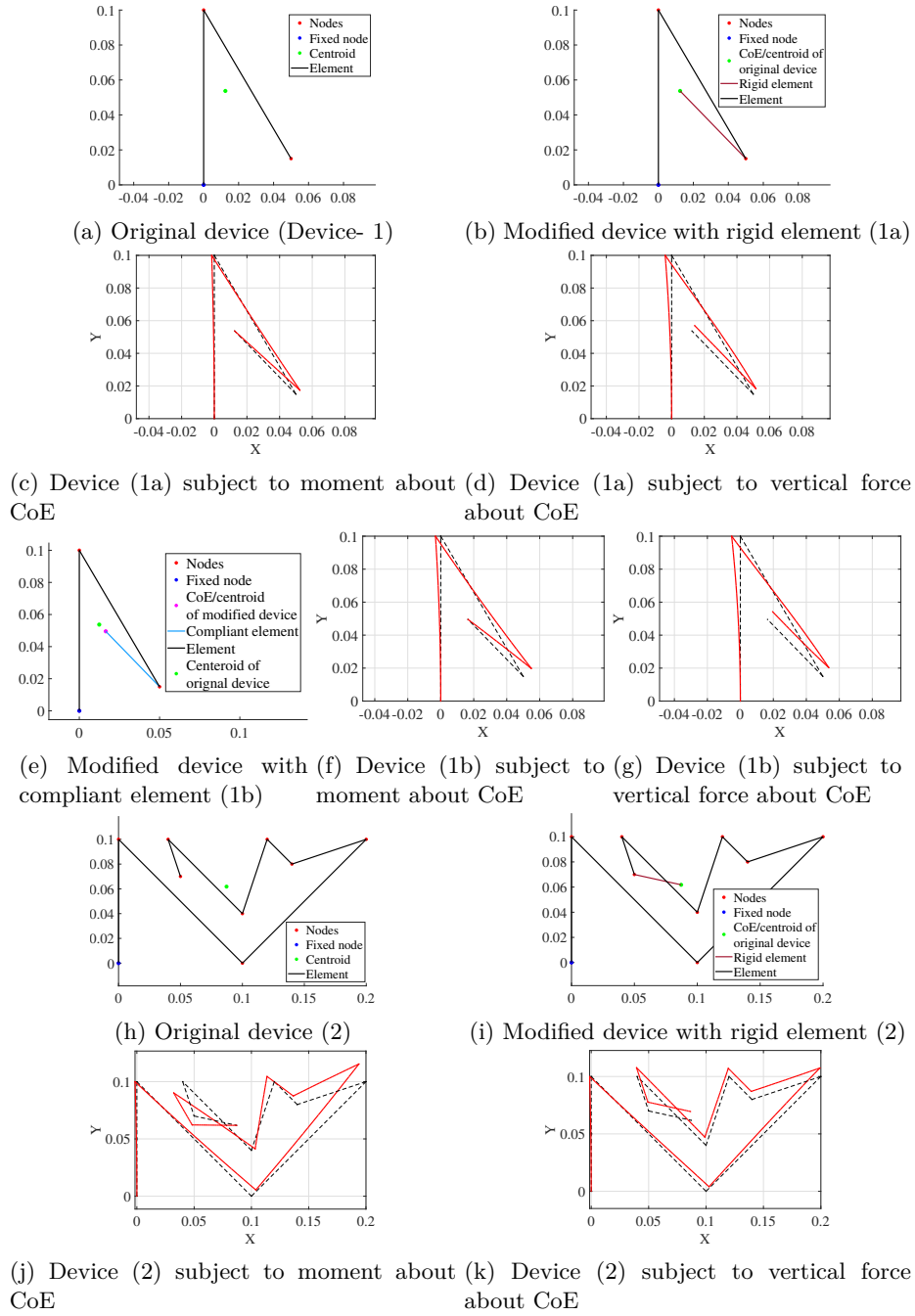


Fig. 4: Examples to illustrate the serial concatenation of beam elements

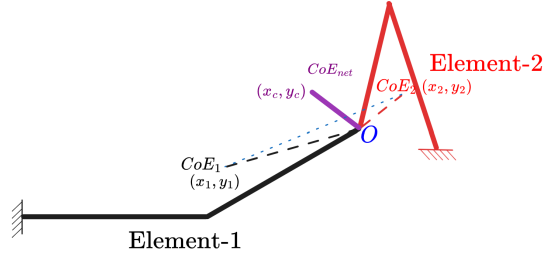


Fig. 5: Illustrative schematic of parallel concatenation of representative elements

3.3 Parallel concatenation of frame elements

It is more convenient to work with a single point stiffness matrix in comparison to a compliance matrix for parallel concatenation. The illustrative schematic for the parallel concatenation of frame elements is shown in Fig. 5.

Let the single point stiffness matrix of element-1 about $\text{CoE}_1 (\{x_1, y_1\})$ and element-2 about $\text{CoE}_2 (\{x_2, y_2\})$ be given by:

$$[\mathbf{K}_1] = \begin{bmatrix} K_{11} & K_{12} & 0 \\ K_{12} & K_{22} & 0 \\ 0 & 0 & K_{33} \end{bmatrix} \quad [\mathbf{K}_2] = \begin{bmatrix} P_{11} & P_{12} & 0 \\ P_{12} & P_{22} & 0 \\ 0 & 0 & P_{33} \end{bmatrix} \quad (14)$$

Rigid-body translation can be performed to relate both matrices about CoE_{net} . As the force is being distributed between two elements and the displacement at the PoI must be the same to satisfy compatibility, the single-point stiffness matrices can be directly summed. Both the elements in entries (3,1) and (3,2) must be set to zero to find the coordinates of CoE_{net} . This leads to the following conditions:

$$\begin{bmatrix} K_{11} + P_{11} & K_{12} + P_{12} \\ K_{12} + P_{12} & K_{22} + P_{22} \end{bmatrix} \begin{Bmatrix} y_{\text{CoE}} \\ -x_{\text{CoE}} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} y_1 \\ -x_1 \end{Bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{Bmatrix} y_2 \\ -x_2 \end{Bmatrix} \quad (15)$$

This is a matrix-weighted average of individual centres of elasticity. In general, a matrix-weighted average will not lie in the line joining the two points.

4 Conclusions

In this work, the concept of the centre of elasticity is elucidated through the stiffness and compliance matrix. In general, a frame can be decomposed into serial and parallel concatenation of beam elements. Relations for finding the centre of elasticity of devices obtained through serial and parallel concatenation of elements are shown here. The centre of elasticity upon parallel concatenation depends on the translation components of the point-stiffness matrix of the elements, which requires extensive computation in comparison to the compliance

associated with the rotational degree of freedom needed for serial concatenation. Also, here it is shown that for a serial concatenation of beam elements of uniform flexural rigidity, the centre of elasticity of the device about the terminal node is coincident with the centroid of the device in its skeletal frame representation. For frames with appropriate symmetry, this insight can aid in the computation of the centre of elasticity solely based on the geometry.

Acknowledgement

The authors acknowledge the synergistic support of all the members of the Multidisciplinary and Multiscale Design and Device (M2D2) Laboratory at the Indian Institute of Science, Bengaluru.

References

1. G. Krishnan, C. Kim, and S. Kota. “An Intrinsic Geometric Framework for the Building Block Synthesis of Single Point Compliant Mechanisms”. In: *Journal of Mechanisms and Robotics* 3 (2010), p. 011001.
2. C. J. Kim. “On the Geometry of Stiffness and Compliance Under Concatenation”. In: *Journal of Mechanisms and Robotics* 12 (2020), p. 021113.
3. O. Sorgonà, O. Giannini, and M. Verotti. “Single-point synthesis of compliant mechanisms”. In: *Precision Engineering* 93 (2025), pp. 58–69.
4. C. J. Kim, Y.-M. Moon, and S. Kota. “A Building Block Approach to the Conceptual Synthesis of Compliant Mechanisms Utilizing Compliance and Stiffness Ellipsoids”. In: *Journal of Mechanical Design* 130 (2008), p. 022308.